

DLQ QUIZ 6-A

① 3bit comparator expressions

$$(A=B) \rightarrow (A_0 \odot B_0)(A_1 \odot B_1)(A_2 \odot B_2)$$

$$(A < B) \rightarrow \bar{A}_2 B_2 + \bar{A}_1 B_1 (A_2 \odot B_2) + \bar{A}_0 B_0 (A_2 \odot B_2)(A_1 \odot B_1)$$

$$(A > B) \rightarrow A_2 \bar{B}_2 + A_1 \bar{B}_1 (A_2 \odot B_2) + A_0 \bar{B}_0 (A_2 \odot B_2)(A_1 \odot B_1)$$

② 2x4 Decoder Expressions

$E_1 \quad \bar{E}_0 \quad \bar{I}_0 \quad \bar{I}_1$

1 1 0 0

1 1 0 1

1 1 1 0

1 1 1 1

$$O_0 = E_1 \bar{E}_0 I_0 I_1$$

$$O_1 = E_1 \bar{E}_0 I_0 \bar{I}_1$$

$$O_2 = E_1 \bar{E}_0 \bar{I}_0 I_1$$

$$O_3 = E_1 \bar{E}_0 \bar{I}_0 \bar{I}_1$$

③ 4x1 Mux Function Implementation $\Sigma(0,1,2,4,5,6)$

$x \quad y \quad z \quad F$

0 0 0 1

0 0 1 1

0 1 0 1

0 1 1 0

1 0 0 1

1 0 1 1

1 1 0 1

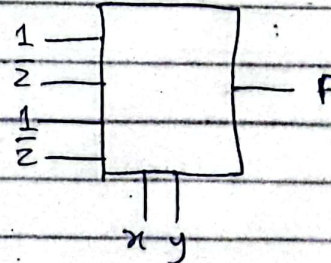
1 1 1 0

$\} F=1$

$\} F=\bar{2}$

$\} F=1$

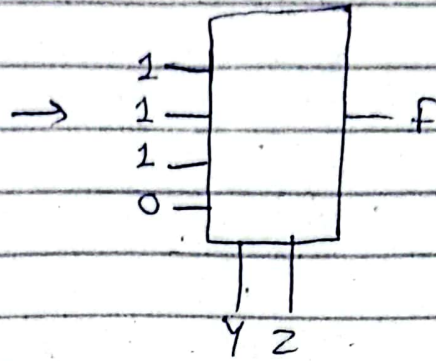
$F=\bar{2}$



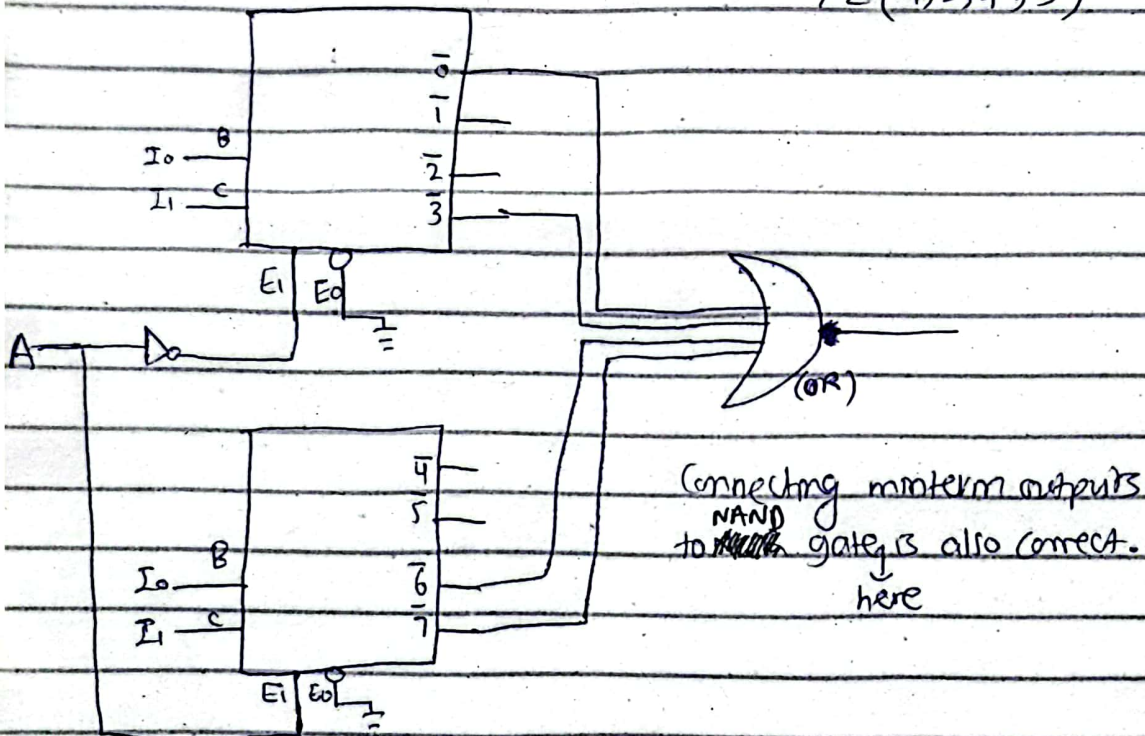
(A)

If x chosen as input

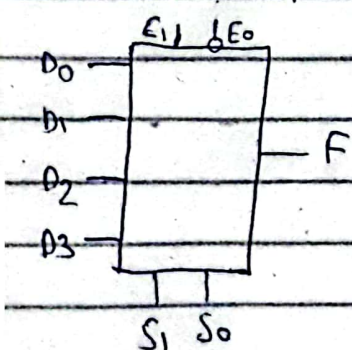
		I_0	I_1	I_2	I_3
0	$\bar{x}$	0	1	0	0
1	x	0	0	1	0
2		1	1	0	0
3		1	0	1	0



⑤ 2x4 Decoders Function Implementation $\Pi(0, 3, 6, 7)$
 $\Sigma(1, 2, 4, 5)$



⑥ 4x1 MUX Expression



S_1	S_0	F
0	0	D_0
0	1	D_1
1	0	D_2
1	1	D_3

Expression

$$F = E_1 \bar{E}_0 \bar{S}_1 \bar{S}_0 D_0 + E_1 \bar{E}_0 \bar{S}_1 S_0 D_1 + E_1 \bar{E}_0 S_1 \bar{S}_0 D_2 + E_1 \bar{E}_0 S_1 S_0 D_3$$

(A)

④ 4x2 Priority Encoder $I_1 < I_3 < I_0 < I_2$
 $\therefore I_2 > I_0 > I_3 > I_1$

I_2	I_0	I_3	I_1	A	B	V
0	0	0	0	x	x	0
1	x	x	x	1	0	1
0	1	x	x	0	0	1
0	0	1	x	1	1	1
0	0	0	1	0	1	1

$I_2 I_0$		$I_3 I_1$			
		00	01	11	10
00	X	0		1	1
01	0	0	0	0	0
11	1	1	1	1	1
10	1	1	1	1	1

for A

$$A = I_2 + \bar{I}_0 I_3$$

$I_2 I_0$		$I_3 I_1$			
		00	01	11	10
00	X	1	1	1	1
01	0	0	0	0	0
11	0	0	0	0	0
10	0	0	0	0	0

for B

$$B = \bar{I}_2 \bar{I}_0$$